

HealthConnect Analytics Testing & Refinement Report

Week 7 — Data Analytics Track

Prepared by: Gladys Ogbeifun

AnalystLab Africa Data Analytics Programme

Prepared following a full round of Week 7 testing on the Week 6 dashboard and KPI outputs. One item — the Data Science cross-track validation — could not be completed this week because no Data Science intern was available to receive and answer the prepared question; this is flagged clearly in Section 8 and carried into Week 8.

1. Executive Summary

Week 7 focused on testing the accuracy of the three new cross-segment dashboard views introduced in Week 6 (reminder status, appointment type × booking lead time, and previous no-show history × booking lead time), together with the headline KPI cards. Each view was independently recomputed from the underlying SQL data and compared against the live Power BI dashboard.

Two of the three cross-segment views initially returned incorrect results — not because the underlying Week 6 analysis was wrong, but because the dashboard measures lacked correct row-level filtering (in one case a leftover measure mix-up, in another a hardcoded filter condition that conflicted with the row context). All three views, along with the KPI cards and a dashboard usability review, passed after the relevant fixes were applied and retested.

Most importantly, the single most significant Week 6 finding — the 72.96% no-show rate among patients with 2+ previous no-shows and a booking lead time of 31+ days — is now confirmed correct at both the raw-data and dashboard level. The one outstanding item is the planned Data Science cross-track validation, which could not be completed this week as no Data Science intern was available to receive the prepared question; this is carried forward as a priority for Week 8.

2. Week 6 → Week 7 Transition

1. Main Week 6 output:

The Week 6 output was the HealthConnect Advanced Analytics & Decision Support Report, including the validated KPI set, the six cross-segment findings, the Data Science handoff, and the improved Power BI dashboard (three new cross-segment views).

2. Most important component or result produced in Week 6:

The most important result was the interaction between previous no-show history and booking lead time — patients with 2+ previous no-shows and a booking lead time of 31+ days recorded a 72.96% no-show rate, the highest segment identified.

3. Main issue, limitation or dependency identified in Week 6:

The Week 6 analysis remained descriptive rather than predictive: relationships were observed but not statistically tested, 90 distance values were missing, and the findings had been handed to the Data Science track for predictive validation but not yet confirmed.

4. Component or workstream that requires testing:

The three new Week 6 cross-segment dashboard views (reminder status, appointment type × booking lead time, and previous no-show history × booking lead time) and the top-level KPI cards.

5. Track(s) relevant to Week 7 testing activities:

Data Analytics (self-testing of the dashboard against the underlying SQL data) and Data Science (an attempted cross-track validation of the handed-off interaction finding, not completed this week).

6. What you intend to test:

Whether each cross-segment dashboard view and the KPI cards reproduce the exact no-show rates already established in the Week 6 SQL analysis, and whether the Data Science track's model treats the previous no-show × lead-time interaction as an important feature.

7. What you expect to improve or validate:

Confirm the dashboard can be trusted to display accurate figures without a user needing to re-run SQL, fix any calculation errors found, and validate with data science whether the highest-risk segment identified in Week 6 carries predictive weight.

3. Part 1 — Testing Readiness

A. Component to Be Tested

What is the Week 6 component? The three new cross-segment dashboard views built in Week 6 — reminder status, appointment type × booking lead time, and previous no-show history × booking lead time — plus the top-level KPI cards (Total Appointments, No-Show Rate, Attended Rate, and Cancelled Rate).

Its intended purpose: To let HealthConnect staff see how no-show risk varies across these segments directly in Power BI, without needing to run SQL to confirm the numbers.

Requirement/objective it should satisfy: Every dashboard value must match the corresponding SQL calculation exactly.

Track(s) connected to this component: Data Analytics (built and tested it); Data Science (intended recipient of the validated interaction finding).

B. Testing Objective

To determine whether the Week 6 dashboard produces KPI and segment results that match the underlying data exactly, and to identify and fix any calculation or display errors before the dashboard is relied on for decision-making.

C. Expected Results / Test Log

Complete one row per test performed. Two example rows are pre-filled below using arithmetic and consistency checks already carried out on the Week 6 figures — add your own rows for dashboard, filter, and cross-track tests.

Test / Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result
Outcome counts reconcile with total records (No-Show + Attended + Cancelled = Total).	$2,423 + 2,314 + 263 = 5,000$	$2,423 + 2,314 + 263 = 5,000$ — matches	Pass	None	None required	N/A
Outcome percentages sum to 100% and match reported rates.	$48.46\% + 46.28\% + 5.26\% = 100.00\%$	$48.46\% + 46.28\% + 5.26\% = 100.00\%$ — matches	Pass	None	None required	N/A

Test / Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result
Previous no-show history sub-rates within the 31+ day band follow the same ordering as the overall marginal rates (0 < 1 < 2+ no-shows).	0 no-shows < 1 no-show < 2+ no-shows in both the marginal table and the 31+ day interaction table	Marginal: 43.51% < 53.49% < 61.02%. Interaction (31+ days): 55.18% < 66.37% < 72.96% — same ordering	Pass	None	None required	N/A
Overall, a 31+ day no-show rate (60.49%) falls within the range implied by its three previous no-show history sub-rates (55.18%–72.96%).	60.49% should sit between 55.18% and 72.96%, consistent with a weighted average.	60.49% falls within the 55.18%–72.96% range.	Pass	None	None required	N/A
The reminder-status dashboard view matches SQL (No = 51.39%, Yes = 47.36%).	No = 51.39%, Yes = 47.36%	1st attempt: 48.46% shown for both (card visual / no row filter context). 2nd attempt: No = 63.64%, Yes = 59.32%, Total = 60.49% (wrong measure applied to visual — the Type × Lead-Time measure was used instead of the Reminder measure).	Fail → Fail → Pass	(1) Visual had no row-level filter context. (2) A correct measure existed, but the wrong measure was placed in the visual's Values field.	Rebuilt the visual as a matrix with reminder_sent on rows; corrected the Values field to the Reminder Test measure.	No = 51.39%, Yes = 47.36% — matches SQL exactly. Pass.
Appointment type dashboard view (31+ days lead time) matches SQL (58.10% / 59.55% / 59.18% / 65.14%).	General Consultation 58.10%, Diagnostic Test 59.55%, Specialist Consultation 59.18%, Follow-up 65.14%	1st attempt: 60.49% shown for every appointment type (the overall 31+ days rate, not split by type).	Fail → Pass	Measure not filtered/split by appointment_type in the visual — same root cause as the reminder test (missing row-level context).	Rebuilt the visual as a matrix with appointment_type on rows and the Type × Lead-Time Test measure in values.	58.10% / 59.55% / 59.18% / 65.14% (Total row: 60.49%) — matches SQL exactly. Pass.
Previous no-show history (0 / 1 / 2+) × 31+ days lead time matches SQL (55.18% / 66.37% / 72.96%).	0 = 55.18%, 1 = 66.37%, 2+ = 72.96%	1st attempt: only the '2+' row displayed a value (72.96%); the '0' and '1' rows were blank.	Fail → Pass	The DAX measure had a hardcoded 'previous_no_shows >= 2' condition inside its FILTER, which conflicted with	Removed the hardcoded previous-no-show condition from the measure, leaving only the booking_lead_days >= 31 filter, so the row-level	All three rows displayed and matched expected values (55.18% / 66.37% /

Test / Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result
				the row context for the '0' and '1' groups (created via a separate grouping column/measurement), causing DIVIDE to return blank for those rows.	Grouping alone determines each row's segment.	72.96%). Pass.
KPI cards (Total Appointments, No-Show Rate, Attended Rate, Cancelled Rate) match reported values with all filters cleared	5,000 / 48.46% / 46.28% / 5.26%	5,000 / 48.46% / 46.28% / 5.26% — matches	Pass	None	None required	N/A
Dashboard usability review — labels, titles, axis clarity, misleading scaling/colour, navigation from headline KPI to high-risk segment	All visuals clearly labelled and readable; no misleading scaling, colour, or sort order; high-risk segment easy to locate	Reviewed all pages, including the three corrected views. No issues identified.	Pass	None	None required	N/A

Rows 1–4 are logic/arithmetic checks performed on the numbers already reported in Week 6. Rows 5–7 are real dashboard tests: three of the three new cross-segment views initially failed for two distinct reasons — missing row-level filter context and a hardcoded filter condition conflicting with row context — and all three passed after correction and retest. Rows 8–9 (KPI cards and usability) passed on first attempt with no changes required.

D. Testing Evidence

This document (with the test log in Section 3C) is the primary evidence record. The following supporting evidence is attached separately via Google Drive:

Dashboard screenshots: the three corrected Matrix visuals (reminder status, appointment type, previous no-show history) and the cleared-filter KPI cards will be attached in the Google Drive folder shared with this submission.

Updated SQL query output: the query results used for each test in Section 3C — the reminder, appointment-type, and previous-no-show grouped queries — will be attached in the Google Drive folder shared with this submission.

Updated Power BI file: the .pbix file (or a PDF export) showing the three rebuilt Matrix visuals will be attached in the Google Drive folder shared with this submission.

4. KPI Validation

The core KPIs from Week 6 have been re-verified arithmetically against the reported counts:

KPI	Reported Value	Recalculation	Result
Total appointments	5,000	2,423 + 2,314 + 263	Matches
No-show rate	48.46%	2,423 / 5,000	Matches
Attended rate	46.28%	2,314 / 5,000	Matches
Cancelled rate	5.26%	263 / 5,000	Matches

Dashboard KPI check: with all filters cleared, the four Card visuals displayed 5,000 / 48.46% / 46.28% / 5.26% — matching the reported values exactly. Pass, no changes required.

Segment-level recalculation: the 72.96% high-risk segment (2+ previous no-shows, 31+ days lead time) was independently recomputed by grouping the raw per-value SQL output (previous_no_shows = 2, 3, 4, 5, summed to 233 total appointments and 170 no-shows) rather than relying on a pre-built grouped query — $170 / 233 = 72.96\%$, confirming the figure from an independent calculation path.

5. Dashboard Testing

Filter testing — completed for all 3 new Week 6 views:

- Reminder-status view: initially returned the same overall value (48.46%, then an incorrect 60.49% - scoped result) for both reminder groups instead of splitting them. Root cause: the visual lacked row-level filter context, and the wrong measure was briefly applied. Rebuilt as a matrix with reminder_sent on rows and the correct measure in values; retest now matches SQL exactly (No = 51.39%, Yes = 47.36%).
- Appointment type view (31+ days lead time): initially returned the same blended value (60.49%) for all four appointment types instead of splitting them. Same root cause as above — missing row-level context. Rebuilt as a matrix with appointment_type on rows; retest now matches SQL exactly across all four types.
- Previous no-show history × booking lead time view (the 72.96% high-risk segment): initially displayed a value for only the '2+' row, with the '0' and '1' rows blank. Root cause: the measure had a hardcoded 'previous_no_shows >= 2' filter that conflicted with the row-level grouping. Fixed by removing that condition from the measure so the grouping column alone determines each row; retest now matches SQL exactly (0 = 55.18%, 1 = 66.37%, 2+ = 72.96%).

Usability review: reviewed all dashboard pages for label clarity, axis readability, misleading scaling or colour, and ease of navigating from the headline 48.46% KPI down to the 72.96% high-risk segment — including the three newly-corrected views. No issues identified; no changes made.

Refined visualisations: all three cross-segment views were corrected from single-value or partially-blank outputs to properly row-filtered Matrix visuals, each now displaying the intended breakdown (reminder status, appointment type, and previous no-show history) rather than one blended or incomplete figure.

6. Validated Findings (Post-Testing)

Findings confirmed:

- Reminder status: the 4.03 percentage-point gap (No = 51.39%, Yes = 47.36%) is confirmed at both the raw-data (SQL) level and, after fixing the dashboard measure, at the dashboard level.
- Appointment type × booking lead time: the ordering and figures (follow-up 65.14% highest, down to General Consultation 58.10%) are confirmed at both the raw-data and (corrected) dashboard level.
- Previous no-show history × booking lead time: the most important Week 6 finding — 2+ previous no-shows and 31+ days lead time producing a 72.96% no-show rate — is now confirmed via an independent recalculation from raw per-value counts ($170/233 = 72.96\%$) and, after fixing the dashboard measure, at the dashboard level (0 = 55.18%, 1 = 66.37%, 2+ = 72.96%).

Findings requiring correction:

No underlying figures required correction. All three discrepancies found were dashboard display/measure issues (missing row-level filter context or a hardcoded filter conflicting with row context), not errors in the original Week 6 analysis or SQL work. Every week 6 figure tested has held up under independent recalculation.

7. Updated Business Insights & Recommendations

Updated insights: testing did not change any of the Week 6 insights — every figure tested reconciled exactly with the original SQL analysis. What changed was confidence in the delivery mechanism: the dashboard can now be trusted to display these figures correctly, which it could not before this week's fixes. The 72.96% high-risk segment in particular is now doubly confirmed (independent recalculation plus corrected dashboard), which strengthens the case for prioritising it operationally.

Updated recommendations: no Week 6 recommendation (Section 9 of the Week 6 report) required substantive revision, since the issues found were dashboard-display bugs rather than analytical errors. One addition is worth making to Recommendation 3 (Investigate High-Risk Appointment Combinations): the underlying data and dashboard now both reliably surface this segment, so HealthConnect staff can be pointed directly to the corrected dashboard view as an operational tool, rather than the finding remaining only in the report.

8. Cross-Track Testing & Validation (HC-POD)

Status: Not completed this week: *The assignment requires at least one meaningful test → finding → action → retest cycle with another track. This was attempted but not completed — see below.*

What was attempted: An outreach message was prepared to send to the Data Science intern, asking whether the previous-no-show × booking lead-time interaction (the 72.96% segment validated in Sections 3 and 6) appears as an important feature in their model in order to close the loop on the Week 6 handoff.

The question intended to be sent to the Data Science intern:

“In your feature importance results, where does the previous-no-show × booking-lead-time interaction rank? I found it's the strongest single segment in the data (72.96% no-show rate) and want to confirm the model is picking it up.”

Why it was not completed: there was no data science intern assigned/available to receive and answer this question this week.

Field	Response
Track collaborated with	Data Science (attempted; not completed)
Project dependency	Data Science's model was expected to test whether the interaction finding handed off in Week 6 (72.96% segment) is a statistically meaningful predictor.
Component/output being tested	The previous no-show × booking lead-time interaction finding
Information/output received	None — no response received
Information/output provided	None — message not sent/answered this week
Testing activity completed	Not completed
Issue or finding identified	No data science intern was assigned/available this week to receive and answer the prepared question
Refinement/action taken	N/A — carried forward to Week 8
Retest result	N/A

Field	Response
What changed as a result	Nothing yet; this remains an open dependency.
Evidence	N/A — to be added once the exchange takes place
How this improved the overall HealthConnect solution	N/A this week — see Section 10 for this being logged as an unresolved dependency going into Week 8.

Per the assignment, a chat or meeting alone would not count as evidence even if one had occurred — it must show Test → Finding → Action → Retest → Validated Improvement. Rather than fabricate that exchange, this section documents honestly that it did not happen this week and is flagged as a priority in Section 10 and Section 11.

9. End-to-End Validation Questions

1. What component did I test? — The Week 6 dashboard's three new cross-segment views (reminder status, appointment type × booking lead time, and previous no-show history × booking lead time) and the top-level KPI cards.
2. What did I expect to happen? — Each view should reproduce the exact no-show rates already established in the underlying SQL analysis.
3. What actually happened? — Two of the three cross-segment views initially displayed incorrect or blended values due to dashboard measure/filter errors, and the third initially displayed correctly for only one of its three rows; the KPI cards and usability review passed on the first attempt.
4. What issue or weakness did I identify? — Three separate measure-construction errors, all traceable to missing or conflicting row-level filter context in the DAX measures behind the visuals.
5. What did I change or improve? — Rebuilt each affected visual as a matrix with the correct grouping field on rows, corrected the measure placed in values where the wrong one had been used, and removed a hardcoded filter condition that was conflicting with row context.
6. What happened when I tested it again? — All three views matched their expected SQL-derived values exactly on retest.
7. Which other track was involved? — None confirmed this week; outreach to the Data Science track was prepared, but no response was received before this document was completed.
8. How does the validated component contribute to HealthConnect? — It gives HealthConnect staff a dashboard they can now trust to correctly surface the highest-risk appointment segments — especially the 72.96% group — without needing someone to re-run SQL to confirm the numbers.
9. What still needs to be completed before Week 8? — The cross-track validation with the Data Science track: confirming whether the previous no-show × lead-time interaction is being picked up as an important feature in their model.

10. Assumptions, Limitations, Risks & Dependencies

Carried forward from Week 6

- Unresolved: the analysis remains descriptive; causal claims have not been established (carried from Week 6, Section 12).
- Unresolved: 90 of 5,000 records (1.8%) have missing distance values; treatment of these was not finalised in Week 6.
- Dependency: the Week 6 findings were handed to the Data Science track for statistical/predictive validation, which had not been confirmed as of Week 6.

New — to complete during Week 7

Issues resolved this week:

- The reminder-status dashboard view was displaying the overall no-show rate instead of splitting by reminder_sent; fixed by rebuilding as a matrix with the correct field and measure.
- The appointment type dashboard view (31+ days) was displaying a single blended rate instead of one rate per appointment type; fixed by rebuilding as a matrix with appointment_type on rows.

New issues discovered during testing:

- Both dashboard errors shared the same root cause — a visual with no (or incorrect) row-level filter context — suggesting the other Week 6 visuals should be spot-checked for the same mistake, including the still-untested previous-no-show × lead-time view.
- A related near-miss: during the reminder-status retest, the correct visual structure was in place, but the wrong measure (Type × Lead-Time Test) was initially dropped into Values, showing how easy it is to mix up similarly-named measures — worth double-checking measure names on every visual, not just the one being actively tested.

New cross-track dependency:

- The Data Science cross-track validation of the 72.96% interaction finding could not be completed this week — no Data Science intern was reachable to answer the prepared outreach question. This is now the primary open dependency going into Week 8.

Technical / data / analytical / modelling limitations found this week:

- No new data or analytical limitations were found beyond those carried forward from Week 6 (missing distance values; descriptive rather than causal analysis). All issues found this week were dashboard/measure construction errors, not data or analytical limitations.

Issues that must be addressed before Week 8:

- Complete the data science cross-track validation of the previous no-show × lead-time interaction finding.
- Attach or submit dashboard screenshots and/or the updated Power BI file as testing evidence, per Section 3D.

11. Week 7 Project Summary

This summary complements, rather than repeats, the sections above.

1. What you planned to test: The three new Week 6 cross-segment dashboard views (reminder status, appointment type × lead time, previous no-show history × lead time) and the top-level KPI cards, plus a cross-track validation with the Data Science intern.

2. What you actually tested: All three cross-segment views and the KPI cards were tested by independently recomputing each in SQL/DAX and comparing to the live dashboard; a dashboard usability review was also completed. The cross-track validation with data science was attempted but not completed.

3. Most important testing results: The dashboard's most important view — the 72.96% high-risk segment (2+ previous no-shows, 31+ days lead time) — was found to be broken (only displaying one of three rows) and was fixed and confirmed correct on retest.

4. Issues or weaknesses identified: Three separate dashboard measure/filter errors, all sharing a root cause of missing or conflicting row-level filter context in the DAX measures.

5. Improvements/refinements made: Rebuilt all three affected visuals as Matrix views with the correct grouping fields on rows; corrected a mismatched measure on one visual; removed a hardcoded filter condition from another that was conflicting with row context.

6. Retesting results: All three corrected views now match their expected SQL-derived values exactly.

7. Track(s) collaborated with: Data Science was the intended collaborator; no intern was available this week to complete the exchange.
8. What was tested collaboratively: Nothing was completed collaboratively this week — see Section 8 for the honest account of what was attempted.
9. What changed as a result: No cross-track change this week; this remains an open item for Week 8.
10. Key findings or validation outcomes: All Week 6 analytical figures held up under independent recalculation. Every discrepancy found was a dashboard construction issue, not an error in the underlying analysis.
11. Major challenges encountered: Diagnosing why a Matrix visual returns a blended or missing value rather than a per-row split — in each case the cause was subtle (a Card visual used instead of a Matrix, a mismatched measure, or a hardcoded filter conflicting with row context) and required checking the DAX itself rather than the visual alone. The absence of an available data science intern was the other main challenge.
12. Important decisions made: Prioritised testing the three new dashboard views and KPI cards over other possible tests, since these were the newest and least-proven Week 6 outputs. Decided to document the incomplete cross-track step honestly rather than fabricate an exchange that did not happen.
13. Remaining limitations: Carried forward from Week 6 — the analysis remains descriptive rather than causal, and 90 distance values remain missing with treatment still undecided.
14. Remaining issues or dependencies: The data science cross-track validation of the 72.96% interaction finding is still outstanding.
15. Your contribution to the overall HealthConnect project: Verified that the analytics layer of HealthConnect—the KPIs and dashboard that other tracks and business stakeholders will rely on—is now confirmed accurate at the segment level, including its single most important risk finding, increasing confidence in the evidence base going into final integration.
16. What must be completed before Week 8: Complete the Data Science cross-track validation; gather and attach dashboard screenshots/updated files as evidence.

12. Submission Checklist

- Track-specific Week 7 output (this document)
- Week 7 Project Summary (Section 11 above)
- Testing & validation evidence (screenshots, SQL output, updated dashboard) — attached via Google Drive; see Section 3D
- HC-POD cross-track evidence (Section 8 above) — not satisfied this week; no Data Science intern was available to complete the exchange
- Supporting files: updated dashboard, notebooks, code, issue log, as applicable